

Project Design Phase Problem – Solution Fit Template

Date	13 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that a problem with the customer has been found and that the solution realised for it actually solves the customer's problem. It helps entrepreneurs, marketers, and corporate innovators identify behavioural patterns and recognise what would work and why.

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behaviour.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behaviour fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.

Template: VBCUA — Problem-Solution Fit Canvas

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) <i>Who is your customer?</i> - Students preparing spoken explanations of academic concepts for graded or self-directed assessment. - Teachers / instructors who need to evaluate multiple students' spoken responses for conceptual understanding.	6. CUSTOMER CONSTRAINTS <i>What constraints prevent customers from taking action or limit their choices?</i> - Instructors have limited time to manually listen to and grade many spoken responses. - Students have no access to a human evaluator when practising outside class hours. - Recording quality (microphone, background noise) can affect transcription accuracy. - Limited familiarity with AI-based assessment tools among some students and staff.	5. AVAILABLE SOLUTIONS <i>Which solutions are available to customers today?</i> - Manual grading: the instructor personally listens to and evaluates each recording (slow, inconsistent across a class). - Text-based quizzes and auto-graders, which test recall but not spoken articulation of a concept. - General-purpose transcription apps, which produce text but do not score it against a reference answer. - Informal peer discussion or study groups for self-assessment (subjective, no objective score).	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS <i>Which jobs-to-be-done (or problems) do you address for your customers?</i>	9. PROBLEM ROOT CAUSE <i>What is the real reason this problem exists?</i> Manual evaluation of spoken responses does not scale with class size or submission volume.	7. BEHAVIOUR <i>What does your customer do to address the problem and get the job done?</i>	

	• Demonstrate and verify genuine understanding of a concept through a spoken explanation. • Evaluate spoken responses consistently, quickly, and at scale against a reference standard.	• No lightweight existing tool combines speech transcription with reference-based semantic scoring for spoken conceptual answers. • Self-assessment of a spoken explanation is inherently difficult without an objective point of comparison.	• Records a spoken explanation using a phone, laptop, or other personal device. • Re-reads notes and rehearses the answer before recording, sometimes taking multiple attempts. • Asks peers or the instructor for informal, unstructured feedback. Indirectly associated: • Reviews the assignment rubric to infer what is expected.	
Identify strong TR & EM	3. TRIGGERS <i>What triggers customers to act?</i> • An upcoming oral assessment, viva, or presentation. • An instructor assigning a “record your explanation” homework task or quiz. • Wanting to self-check understanding before a graded exam.	10. YOUR SOLUTION <i>If you are working on an existing solution, write down your current solution first.</i> • VBCUA transcribes a student's spoken explanation using OpenAI Whisper, compares it semantically to a stored reference answer using Sentence Transformers, and returns an instant percentage score, grade tier (Excellent / Good / Fair / Needs Improvement), and written feedback. • A downloadable PDF report captures the transcript, score, grade, and feedback for the student's or evaluator's records. • This removes the need for an instructor to manually listen to and grade every recording for a first-pass assessment, while giving students immediate, objective feedback.	8. CHANNELS of BEHAVIOUR <i>What kind of actions do customers take online / offline?</i> 8.1 Online • Course LMS submission portals. • Email submission of audio files to instructors. • General-purpose audio recording and transcription apps. 8.2 Offline • In-class oral presentations, vivas, or spoken quizzes. • Face-to-face office hours with the instructor.	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER <i>How do customers feel when they face a problem or job, and afterwards?</i> • Before: anxious and uncertain about their own understanding; unsure how a spoken answer will be judged. • After (using VBCUA): reassured by an objective score, more in control of their preparation, and motivated to re-practise. • For evaluators: relieved of repetitive manual grading and more confident in consistent scoring across students.			

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>